1

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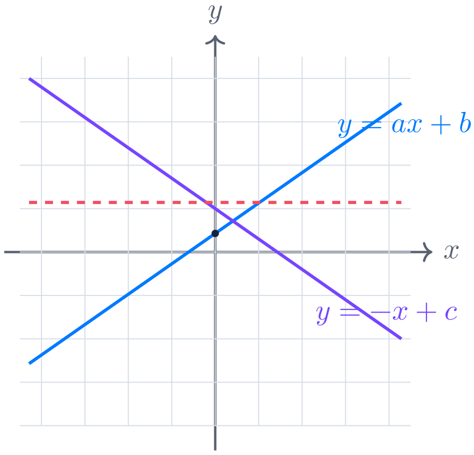


1.
2.
3. xOa

1 1

a,

$$\begin{cases} \dfrac{(y^2 - xy + 2x - 2y)\sqrt{x+1}}{\sqrt{6-x}} = 0, \\ x + y - a - 1 = 0 \end{cases}$$

.

$$\frac{A}{B} = 0 \iff \begin{cases} A = 0, \\ B \neq 0, \end{cases} \qquad \sqrt{A} \qquad \iff A \geq 0, \qquad \frac{1}{\sqrt{A}} \qquad \iff A > 0.$$

:

$$\begin{cases} (y^2 - xy + 2x - 2y)\sqrt{x+1} = 0, \\ 6 - x > 0. \end{cases}$$

$$y^2 - xy + 2x - 2y = (y-2)(y-x),$$

$$(y-2)(y-x)\sqrt{x+1} = 0.$$

$$6 - x > 0 \iff x < 6, \qquad x + 1 \geq 0 \iff x \geq -1.$$

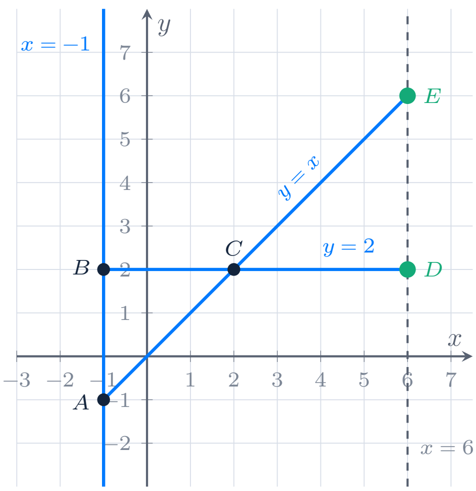
,

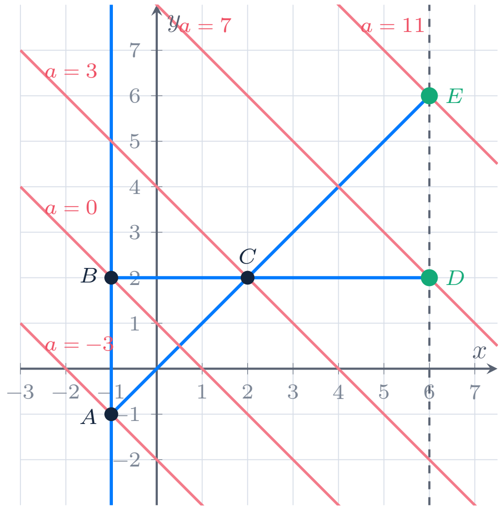
$$\boxed{\begin{cases} -1 \leq x < 6, \\ \left[\begin{array}{l} y = 2, \\ y = x, \\ x = -1. \end{array}\right. \end{cases}}$$

:

$$x + y - a - 1 = 0 \iff \boxed{y = -x + a + 1}.$$

$k = -1$, $(0; a+1)$.





$y = -x + a + 1$:

$$\begin{aligned}
A(-1;-1):&\quad -1 = 1 + a + 1 &&\Rightarrow a = -3,\\
B(-1;2):&\quad 2 = 1 + a + 1 &&\Rightarrow a = 0,\\
C(2;2):&\quad 2 = -2 + a + 1 &&\Rightarrow a = 3,\\
D(6;2):&\quad 2 = -6 + a + 1 &&\Rightarrow a = 7,\\
E(6;6):&\quad 6 = -6 + a + 1 &&\Rightarrow a = 11.
\end{aligned}$$

$a \le -3$	1	,
$-3 < a \le 0$	2	,
$0 < a < 3$	3	,
$a = 3$	2	,
$3 < a < 7$	3	,
$7 \le a < 11$	2	,
$a \ge 11$	1	.

: $a \in (-3;0] \cup \{3\} \cup [7;11)$**.**

2 2

a,

$$\begin{cases} y^2 - 10y - x^2 + 6x + 16 = 0, \\ y - ax = 1, \\ x > 1 \end{cases}$$

.

:

$$y^2 - 10y + 25 = (y-5)^2, \qquad x^2 - 6x + 9 = (x-3)^2.$$

$$(y-5)^2 - (x-3)^2 = 0,$$

$$(y - 5 - (x-3))(y - 5 + (x-3)) = 0.$$

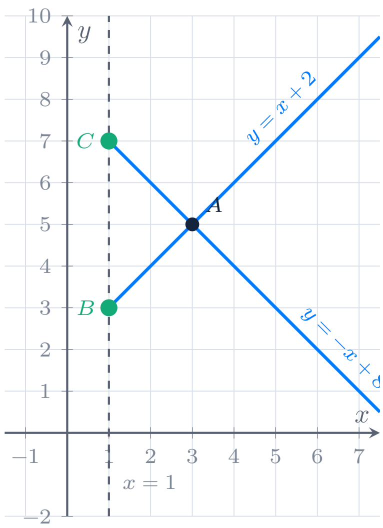
$$\left[\begin{aligned} &y = x + 2, \\ &y = -x + 8. \end{aligned}\right.$$

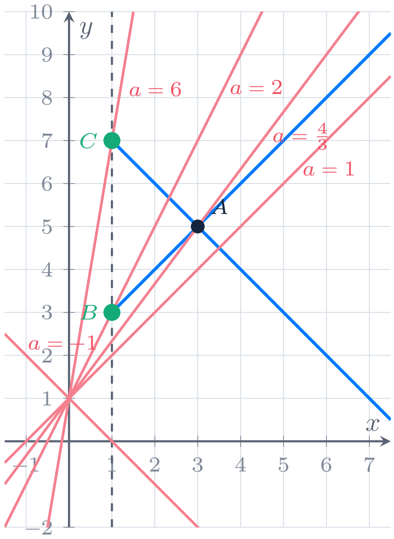
:

$$y - ax = 1 \iff \boxed{y = ax + 1}.$$

,

$$\begin{cases} \left[\begin{aligned} &y = x + 2, \\ &y = -x + 8, \end{aligned}\right. \\ y = ax + 1, \\ x > 1. \end{cases}$$





$$
\begin{array}{lll}
1) & y = ax + 1 \parallel y = -x + 8 & \Rightarrow a = -1, \\
2) & y = ax + 1 \parallel y = x + 2 & \Rightarrow a = 1, \\
3) & A: \; x + 2 = -x + 8 \Rightarrow x = 3, \; y = 5, & \\
 & \quad 5 = 3a + 1 & \Rightarrow a = \dfrac{4}{3}, \\
4) & B(1;3): \quad 3 = a + 1 & \Rightarrow a = 2, \\
5) & C(1;7): \quad 7 = a + 1 & \Rightarrow a = 6.
\end{array}
$$

$a \le -1$	0	,
$-1 < a \le 1$	1	,
$1 < a < \frac{4}{3}$	2	,
$a = \frac{4}{3}$	1	,
$\frac{4}{3} < a < 2$	2	,
$2 \le a < 6$	1	,
$a \ge 6$	0	.

: $a \in (-1; 1] \cup \left\{\frac{4}{3}\right\} \cup [2; 6)$**.**

3 3

a,

$$\frac{4x^2 - a^2}{x^2 + 6x + 9 - a^2} = 0$$

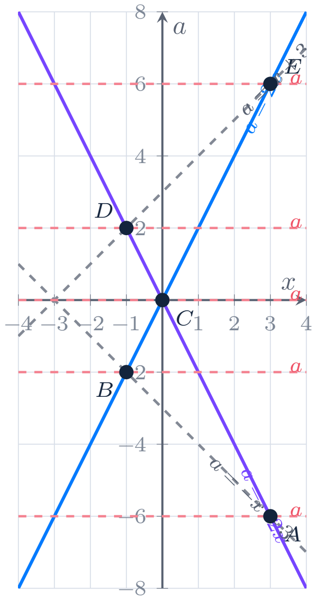
.

$$\frac{A}{B} = 0 \iff \begin{cases} A = 0, \\ B \neq 0. \end{cases}$$

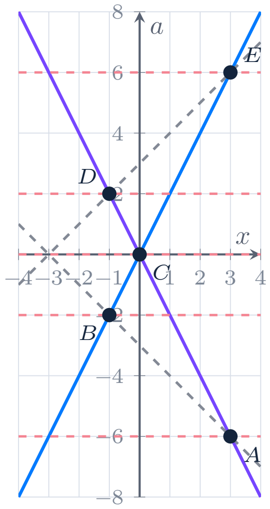
$$\begin{cases} 4x^2 - a^2 = 0, \\ x^2 + 6x + 9 - a^2 \neq 0 \end{cases} \iff \begin{cases} (2x - a)(2x + a) = 0, \\ (x + 3 - a)(x + 3 + a) \neq 0. \end{cases}$$

xOa:

$$\boxed{\begin{cases} \left[\begin{array}{l} a = 2x, \\ a = -2x, \end{array}\right. \\ a \neq x + 3, \\ a \neq -x - 3. \end{cases}}$$



$$
\begin{array}{lll}
A: & a=-2x,\ a=-x-3 & \Rightarrow -2x=-x-3 \Rightarrow x=3,\ a=-6,\\
B: & a=2x,\ a=-x-3 & \Rightarrow 2x=-x-3 \Rightarrow x=-1,\ a=-2,\\
C: & a=2x,\ a=-2x & \Rightarrow x=0,\ a=0,\\
D: & a=-2x,\ a=x+3 & \Rightarrow -2x=x+3 \Rightarrow x=-1,\ a=2,\\
E: & a=2x,\ a=x+3 & \Rightarrow 2x=x+3 \Rightarrow x=3,\ a=6.
\end{array}
$$



$$a \in \{-6; -2; 0; 2; 6\}.$$

$a = a_0$.

: $a \in \{-6; -2; 0; 2; 6\}$**.**